Transcriptomic Analysis Report: Alzheimer's Disease vs Old Control

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Data Source: GSE153873 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE153873)

1. Introduction and Data Description

The dataset GSE153873 consists of RNA-seq profiles from human post-mortem hippocampal tissue (lateral temporal lobe). The study aims to identify transcriptomic and epigenetic alterations associated with Alzheimer's Disease (AD).
For this project, I performed a differential expression analysis comparing:

AD Group: Samples affected by Alzheimer's Disease.

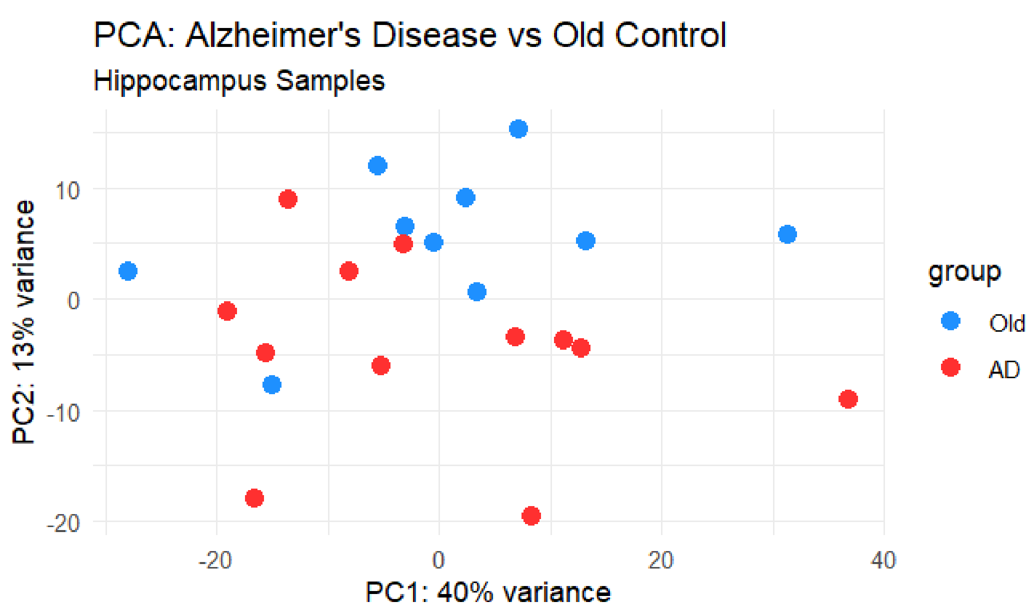
Old Control: Age-matched healthy individuals.

The analysis was performed using the DESeq2 package in R. Raw counts were filtered (minimum 10 reads in at least 3 samples) and normalized using Variance Stabilizing Transformation (VST).

2. Exploratory Data Analysis (PCA)

The PCA plot (PC1: 40% variance) shows that the samples from the "Old" and "AD" groups generally tend to cluster separately, although there is some overlap.

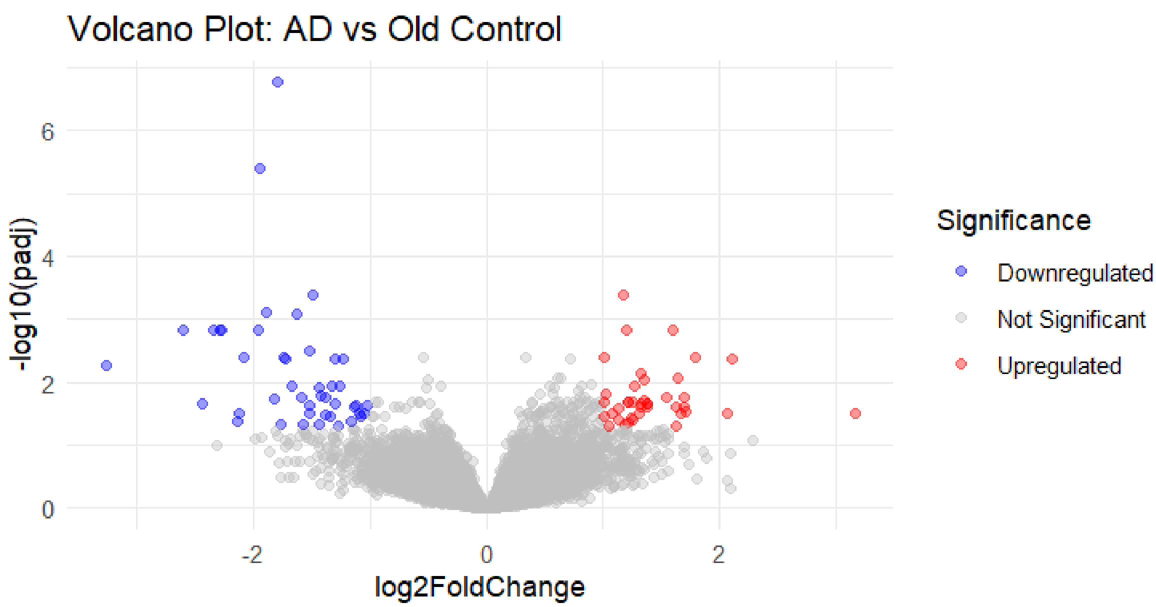
This overlap is expected in human post-mortem brain studies due to high biological heterogeneity and varying stages of disease progression. However, PC1 successfully captures the major transcriptomic shift between healthy aging and the AD state.



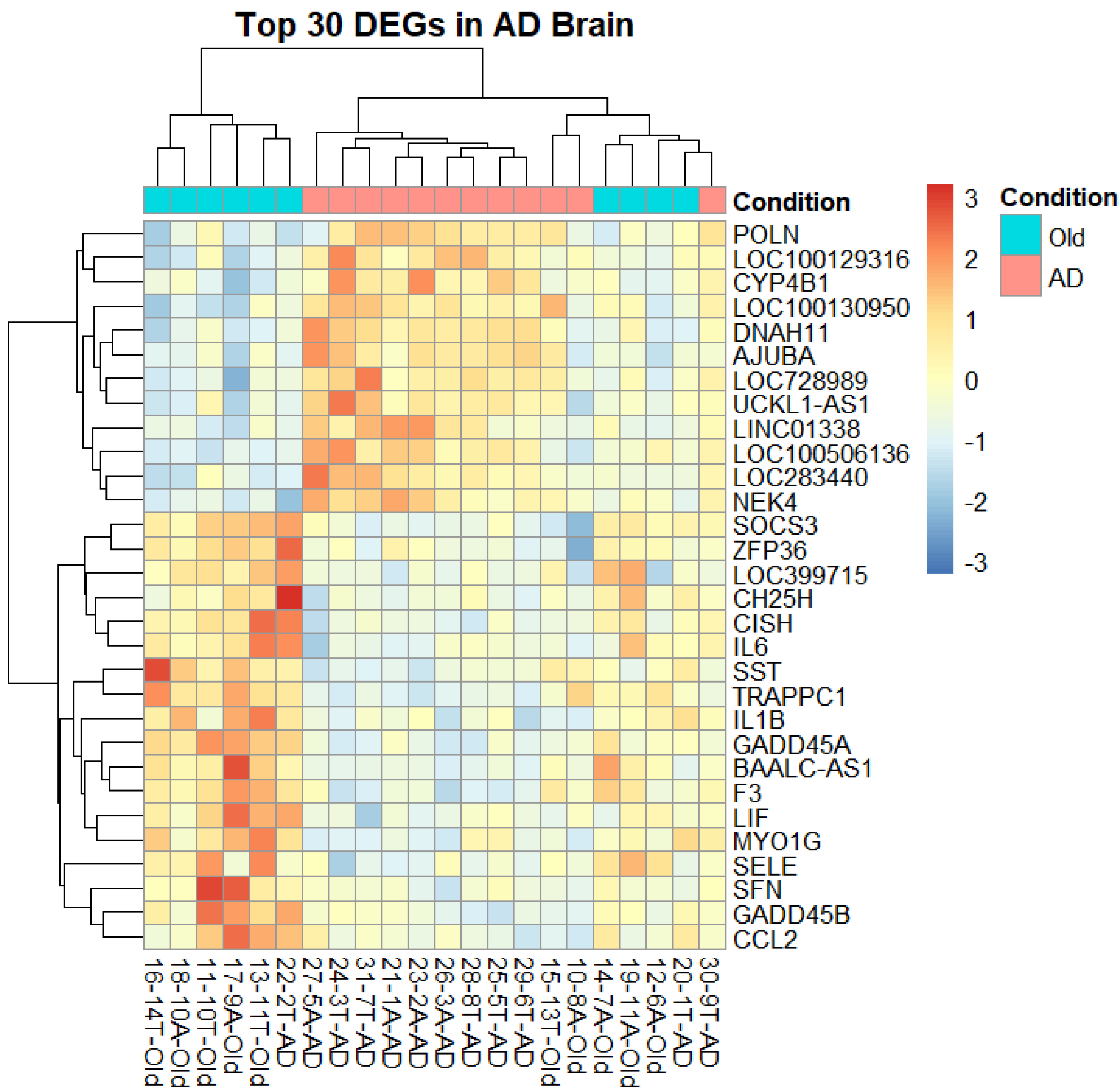
3. Differential Gene Expression Analysis (DEGs)

I identified significant DEGs using a threshold of |log2FoldChange| > 1 and padj < 0.05.

Volcano Plot: Shows a significant number of both up-regulated (red) and down-regulated (blue) genes.



Heatmap: The top 30 DEGs clearly distinguish the two groups. Notably, several pro-inflammatory markers and cytokines are highly up-regulated in AD samples (e.g., IL1B, IL6, CCL2, SOCS3). This suggests a strong neuroinflammatory component in the hippocampal tissue of AD patients.



4. Enrichment Analysis (GO & KEGG)

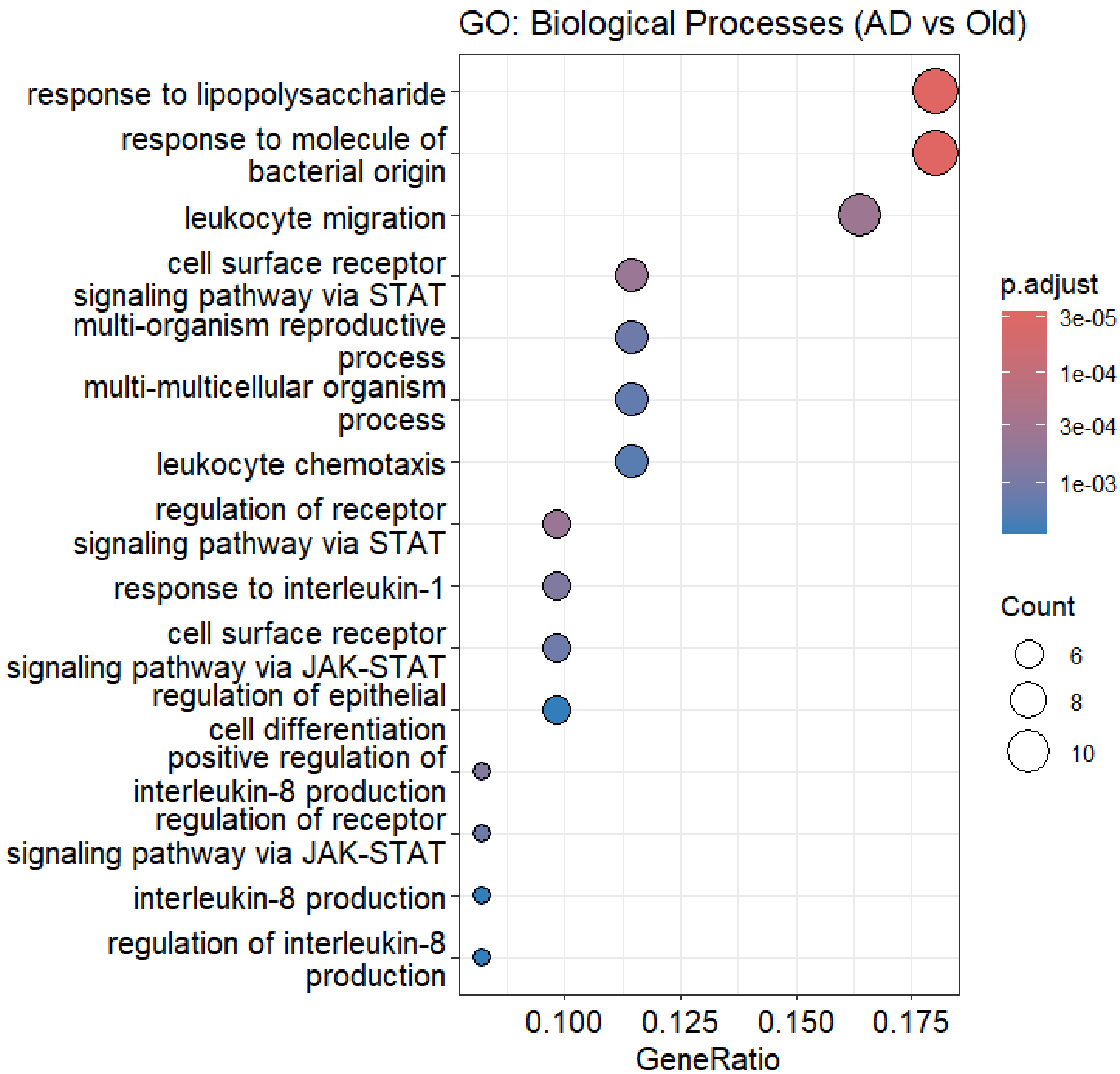
GO Biological Processes:

The most enriched terms are heavily related to the immune response:

Response to lipopolysaccharide and molecule of bacterial origin.

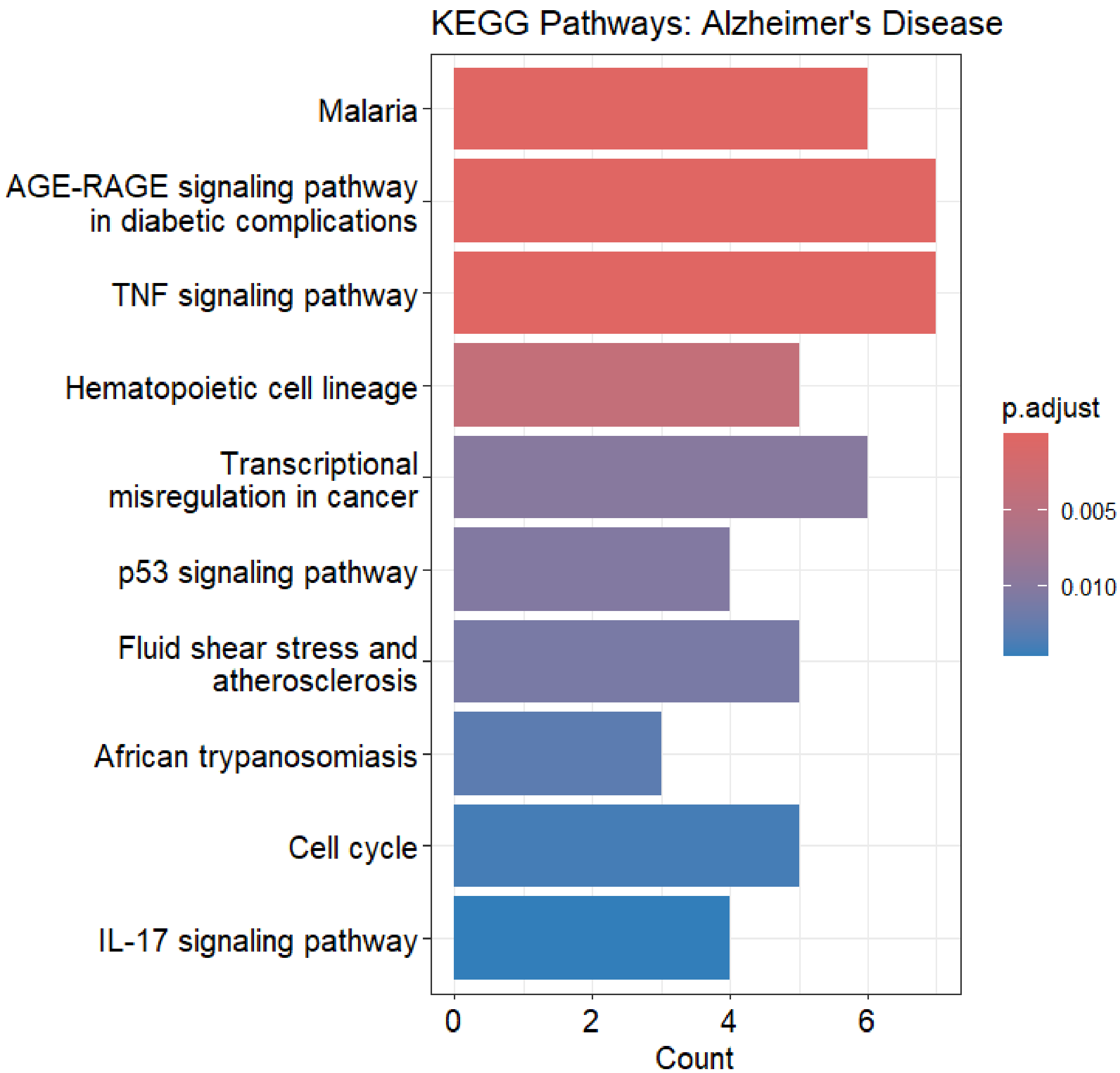
Leukocyte migration and chemotaxis.

JAK-STAT and interleukin-8 production.



KEGG Pathways:

While the "Alzheimer disease" pathway was expected, we see several "infectious" pathways like Malaria and African trypanosomiasis.

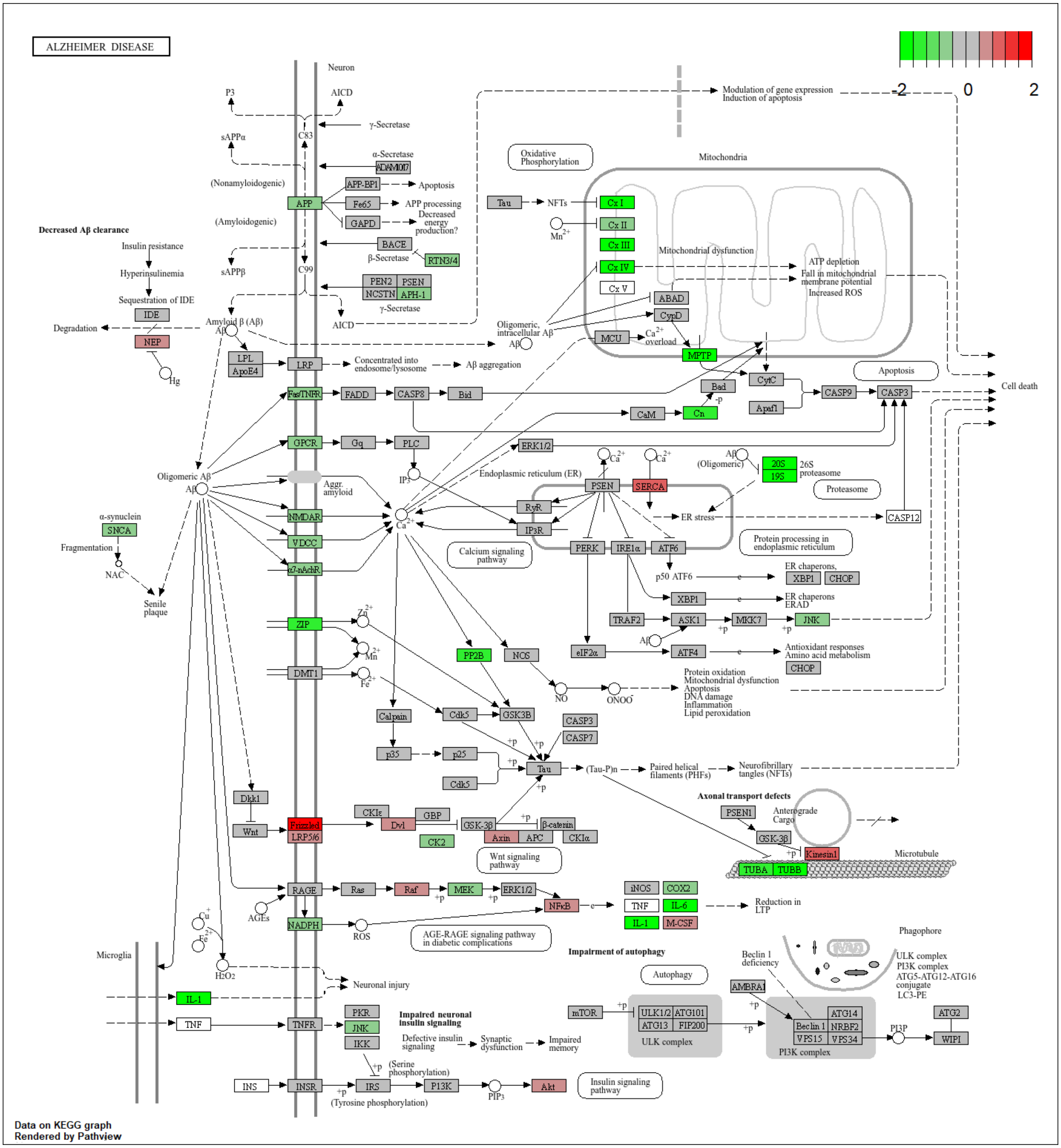


Unexpected findings: The appearance of the "Malaria" pathway in a brain study might seem strange. However, in bioinformatics, this usually happens because AD and Malaria share common inflammatory signaling cascades, such as the activation of the TNF signaling pathway and the AGE-RAGE signaling pathway in diabetic complications (also seen in the results).

5. Pathway Visualization and Biological Assumption

Using the pathview package, I mapped the results onto the KEGG Alzheimer Disease pathway (hsa05010).

Observations: There is a clear up-regulation of genes involved in neuroinflammation (TNF, IL-1) and potentially apoptosis. Interestingly, some components related to mitochondrial function and the electron transport chain (CX I-V) show mixed or down-regulated signals, which aligns with the "mitochondrial dysfunction" hypothesis in AD.



Biological Assumption:

Based on the integrated results, I hypothesize that the transcriptomic signature in this AD cohort is dominated by sterile neuroinflammation. The enrichment of leukocyte migration and cytokine signaling (IL-1, IL-6, TNF) suggests that the disease progression is significantly driven by microglial activation and an overactive innate immune response, rather than just the accumulation of Amyloid-beta and Tau proteins. The "Malaria" hit in KEGG reinforces that the inflammatory "cytokine storm" in the AD brain mimics the systemic inflammation seen in severe infections.